ARVIN BOZORGI

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Education

- *McDonogh High School,* Expected Graduation: 2027
- Dean's List 2024, '25, '26
- 2026 Computer Science Award recipient - Presented to an individual for outstanding achievement and interest in computer science
- 35 ACT

Experience

UMD Machine Learning Internship, *Intern* *June 2026 - Present*

- Paid in-person internship at the University of Maryland working under the guidance of Professor Ohadi and his team to predict energy consumption of buildings
- Programmed a script to automate data cleaning for training ML/AI prediction models

UMD Hackathon *February 2026*

- Competed in a Hackathon hosted by UMD, representing and recognized for my quick problem-solving and my ability to collaborate with a team.

FRC Team, *Programmer for McDonogh School* *January 2024 - Present*

- Programmer for the school's FRC team, developing robot control software in Java using WPILib and other FRC libraries for competition robots.

Glaucoma Classification Research, *Intern* *June 2025 - Present*

- Drafting research paper, "Impact of Different Data Augmentation Techniques for Glaucoma Detection Using a SWIN Transformer," with Dr. Javidian for publication.
- Annotated 1,500+ retinal images in Roboflow and trained a model to isolate optic nerve regions for focused glaucoma detection.
- Building PyTorch classification models and testing augmentation strategies to address dataset imbalance while tracking metrics in Weights & Biases.

Conceptbytes Team, *Developer [Demonstration]* *August 2024 - Present*

- Collaborate with a well-known influencer and engineer to develop an innovative AI project featured on a professional platform, gaining significant attention.
- Develop computer vision systems using Python, MediaPipe, and OpenCV.

A Byte For All, *Developer & Project Branch Director* *June 2025 - Apr 2026*

- Lead a team of student developers through community projects, such as web development.
- Create websites (HTML, CSS, JS) for nonprofits, businesses, and school clubs.

High School STEM Center, *Tutor* *August 2025 - Present*

- Selected through a competitive interview based on STEM coursework and soft skills to tutor students during school hours.

Projects

Computer Vision Chess Glasses *March 2025*

- Trained a model to detect chessboard corners and warp live footage for analysis.
- Tracked piece movement using binary thresholding and canny edge detection.
- Streamed video from Meta Raybans and displayed moves in real time with PyGame.

Soccer Tic-tac-toe *[Demonstration]* *December 2024*

- Hand-annotated 4,000+ images to train a computer vision model that tracks a soccer ball in real time. Used this to project a PyGame tic-tac-toe board onto a wall and detected ball impact location to register moves.
- Created an alternative (projector + laptop + webcam) to $5,000 commercial systems.

Warfields II Home Owners Association Website *[Website]* *June 2025*

- Developed a modern and accessible HOA website based on board member feedback.

Jarvis (Hand-Tracking Projected UI), *[Demonstration]* *August 2024 - Present*

- Built a menu system and two arcade apps (Space Invaders, Brick Breaker) for Jarvis, an open-source Iron Man–style projected UI using OpenCV, MediaPipe, and PyGame.
- Sold apps to the project founder for $200; featured on his Instagram page.
- Joined the official GitHub developer team and remain an ongoing contributor.

Leadership

Competitive Programming Club, *President & Founder* *May 2025 - Present*

- Lead discussions on competition strategy and coordinate teams to represent our school in regional programming contests.

Founder of Machine Learning Club, *President & Founder* *May 2025 - Present*

- Taught neural network programming and machine learning applications.
- Create presentations and Python code walkthroughs for beginner-level models.

Founder of 3D Design Club *August 2024 - May 2025*

- Founded and led a CAD and 3D printing club, teaching Fusion 360 and prototyping.

Technical Skills:

- **Programming**: Python, Java, C, C#, C++, HTML, JavaScript, CSS, Lua
- **AI & Machine Learning**: TensorFlow, PyTorch, OpenCV, MediaPipe, Weights & Biases, Roboflow, Neural networks, facial recognition, hand tracking
- **3D Design & Prototyping**: Fusion 360, Blender modeling, 3D Printing
- **Certificates:** PCEP – Certified Entry-Level Python Programmer, HarvardX CS50